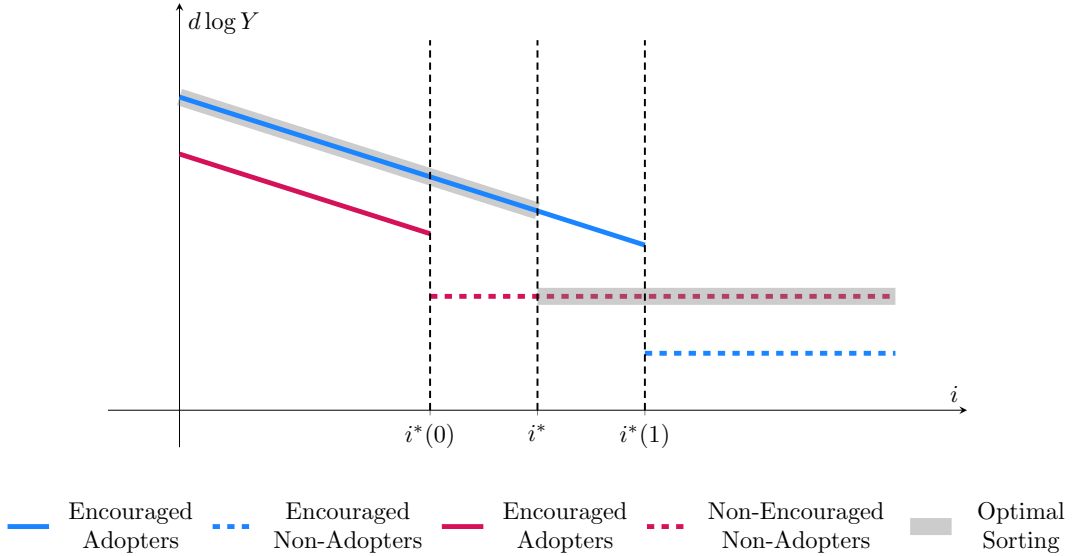


patterns: workers with  $i < i^*$  sort into encouraged firms and adopt chatbots, while those with  $i > i^*$  sort into non-encouraged firms and do not adopt.

Figure A.3 illustrates the case where employer encouragement increases adopter earnings. However, as shown above, encouragement can also reduce adopter earnings—reversing the solid lines—depending on model parameters. In fact, if employer-encouraged chatbot use provides enough amenity value (i.e., if  $\alpha_{c1}$  is low), some encouraged adopters may even experience earnings losses relative to non-encouraged non-adopters, with the solid blue line crossing the dashed red one.

Figure A.3: The Impact of AI Chatbots on Worker Sorting, Adoption, and Earnings



*Notes:* This figure illustrates how AI chatbots affect workers' sorting, adoption, and earnings across the distribution of AI advantage. The x-axis ranks workers by their chatbot advantage  $i$ ; the y-axis shows the resulting impact on earnings. The blue and red lines represent earnings effects for each combination of worker and firm types. The gray line highlights the optimal sorting and adoption patterns.

**Workplace employment.** The effect of employer encouragement on workplace employment (measured in headcounts,  $N$ ) depends on whether the increased demand for human capital is offset by the productivity gains from adoption.

The increase in human capital demand is given by:

$$\log H_1 - \log H_0 = \frac{\epsilon}{\epsilon - 1} \kappa. \quad (33)$$

The average productivity gain among encouraged adopters is given by:

$$\mathbb{E}[B_{ij}(\alpha) \mid i \leq i^*, E_j = 1] = (1 - \gamma)(\alpha_{b0} + \alpha_{b1}) + \gamma(\alpha_{c0} + \alpha_{c1} - (\log W_1 - \log W_0)) \quad (34)$$

$$= (1 - \gamma)(\alpha_{b0} + \alpha_{b1}) + \gamma(\alpha_{c0} + \alpha_{c1} + \kappa/(\epsilon - 1)) \quad (35)$$

where  $\gamma = \frac{\beta_b}{2\beta} \geq 0$ . Equation (34) generalizes Equation (8) to account for worker sorting across firms.

Combining Equations (33) and (35), the effect of employer encouragement on workplace employment is:

$$\log N_1 - \log N_0 = \frac{\epsilon}{\epsilon - 1} \kappa - (1 - \gamma)(\alpha_{b0} + \alpha_{b1}) - \gamma(\alpha_{c0} + \alpha_{c1} + \kappa/(\epsilon - 1)) \quad (36)$$

$$= \frac{\epsilon - \gamma}{\epsilon - 1} \kappa - (1 - \gamma)(\alpha_{b0} + \alpha_{b1}) - \gamma(\alpha_{c0} + \alpha_{c1}) \quad (37)$$

In particular, if encouragement is relatively costless ( $\kappa$  is low), product demand is elastic ( $\epsilon$  is high), workers are relatively homogeneous in their individual chatbot productivity ( $\gamma$  is low), and chatbots offer large productivity gains ( $\alpha_b$  is high) but are costly to adopt ( $\alpha_c$  is high), then employer encouragement will tend to reduce workplace employment.

**Workplace wage bills.** Encouraged firms increase their wage bills by:

$$\log(W_1 H_1) - \log(W_0 H_0) = (1 - \epsilon)(\log W_1 - \log W_0) = \kappa. \quad (38)$$

The prediction that employer encouragement unambiguously raises wage bills is a consequence of modeling AI chatbots solely as tools that augment worker productivity (see Equation (13)). As discussed in Section A.2.1, this result breaks for more general production technologies.

### A.2.5 Empirical Identification

This section demonstrates that the causal effects of AI chatbots on worker and workplace outcomes can be identified using a series of difference-in-differences estimators. We further use the model to structurally interpret these estimates.

The analytical derivations above, for simplicity, assumed that differences across workers and firms arise solely from chatbot use and employer encouragement. A more realistic setting allows for baseline heterogeneity in worker productivity ( $H_{ij}(A_{ij}) = H_{i0} \exp(B_{ij}A_{ij})$ ) and firm-level product demand ( $Y_j = Y_{j0}P_j^{-\epsilon}$ ). For example, Figure 2.(a) shows that workers who adopted AI chatbots were already earning more before chatbots became available, and that employers who adopted enterprise solutions were already larger prior to adoption.

Even with such heterogeneity, the causal effects of chatbot adoption can still be identified using difference-in-differences. The key assumption is that baseline heterogeneity is time-invariant and enters the outcome equations in a log-additive way, such that it cancels out when examining log changes over time. For example:

$$\Delta H_{ij} = \log(H_{i0} \exp(B_{ij}A_{ij}^{post})) - \log(H_{i0} \exp(B_{ij}A_{ij}^{pre})) = B_{ij}(A_{ij}^{post} - A_{ij}^{pre}) \quad (39)$$

$$\Delta Y_j = -\epsilon(\log(Y_{j0}P_j^{post}) - \log(Y_{j0}P_j^{pre})) = -\epsilon(\log P_j^{post} - \log P_j^{pre}), \quad (40)$$

where  $\Delta X = \log X^{post} - \log X^{pre}$  denotes log changes over time (after vs. before the arrival of AI chatbots).

More generally, if baseline heterogeneity varies over time, the identifying assumption is that workers (firms) do not adopt (encourage) chatbots in response to anticipated shocks to their baseline human capital  $H_{i0}$  (or product demand  $Y_{j0}$ ). This assumption is consistent with the policy rules for adoption and encouragement (Equations (16) and (22)), which do not depend on baseline human capital or product demand.

**Worker-level adoption–encouragement difference-in-differences.** The earnings equation in (28) motivates the three difference-in-differences estimators used in Section E.2.1.

### 1. Productivity effect among non-encouraged adopters (*Non-encouraged adopters*)

vs. non-encouraged non-adopters, before vs. after AI)

$$\begin{aligned}\Delta_{\text{NE,A} - \text{NE,NA}}^Y &= \mathbb{E}[\Delta \log Y_{ij} \mid A_{ij} = 1, E_j = 0] - \mathbb{E}[\Delta \log Y_{ij} \mid A_{ij} = 0, E_j = 0] \\ &= \gamma \alpha_{c0} + (1 - \gamma) \alpha_{b0},\end{aligned}\tag{41}$$

where  $\gamma = \frac{\beta_b}{2\beta} \geq 0$  (see Equation (8)).<sup>23</sup>

## 2. Differential skill–price effect in encouraged firms (*Encouraged non-adopters vs. non-encouraged non-adopters, before vs. after AI*)

$$\begin{aligned}\Delta_{\text{E,NA} - \text{NE,NA}}^Y &= \mathbb{E}[\Delta \log Y_{ij} \mid A_{ij} = 0, E_j = 1] - \mathbb{E}[\Delta \log Y_{ij} \mid A_{ij} = 0, E_j = 0] \\ &= \mathbb{E}[d \log W_j \mid E_j = 1] - \mathbb{E}[d \log W_j \mid E_j = 0] \\ &= -\kappa/(\epsilon - 1).\end{aligned}\tag{42}$$

## 3. Combined effects for encouraged adopters (*Encouraged adopters vs. non-encouraged non-adopters, before vs. after AI*)

$$\begin{aligned}\Delta_{\text{E,A} - \text{NE,NA}}^Y &= \mathbb{E}[\Delta \log Y_{ij} \mid A_{ij} = 1, E_j = 1] - \mathbb{E}[\Delta \log Y_{ij} \mid A_{ij} = 0, E_j = 0] \\ &= -\kappa/(\epsilon - 1) + \gamma(\alpha_{c0} + \alpha_{c1}) + (1 - \gamma)(\alpha_{b0} + \alpha_{b1}).\end{aligned}\tag{43}$$

By contrast, when workers are perfectly mobile, Equation (35) implies that:

$$\Delta_{\text{E,A} - \text{NE,NA}}^Y = -\kappa/(\epsilon - 1) + \gamma(\alpha_{c0} + \alpha_{c1} + \kappa/(\epsilon - 1)) + (1 - \gamma)(\alpha_{b0} + \alpha_{b1}).\tag{44}$$

**Workplace-level difference-in-differences.** Section 3.3 analyzes the difference-in-differences of employer encouragement on workplace wage bills and employment. Following Section A.2.4, these estimates identify:

### 1. Workplace employment effect of employer encouragement. Following Equation (37), the difference-in-differences estimate of employer encouragement on workplace

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<sup>23</sup>The frictionless sorting model (Equation (26)) predicts that all adopters sort into encouraged firms, while all non-adopters sort into non-encouraged ones, leaving no workers left to identify Equation (41). In practice, however, Section 3 finds little evidence of worker resorting in response to chatbot adoption. For this reason, Equation (41) derives the structural interpretation without worker mobility.